Data Models & Relational Model

Mirunalini.P

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Session Objective

- To learn about Data models
- To learn about Relational databases

Session Outcome

At the end of this session, participants will be able to:

- Understand the different data models
- Understand Relational database Basic

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Data Abstraction

One fundamental characteristic of the database approach is that it provides some level of data abstraction.

- Refers to the suppression of details of data organization and storage.
- Highlighting the essential features for improved understanding of data.
- DB approach supports data abstraction so that different users perceive data at their preferred level of detail.

Data Model

A **data model** is a collection of concepts that can be used to describe the structure of a database—provides the necessary means to achieve this abstraction

- Structure of a database - Data types, relationships and constraints that apply to the data.
- Basic Operations for manipulating these structures
- Valid user-defined operations that are allowed on the database objects.

Data Model Structure and Constraints:

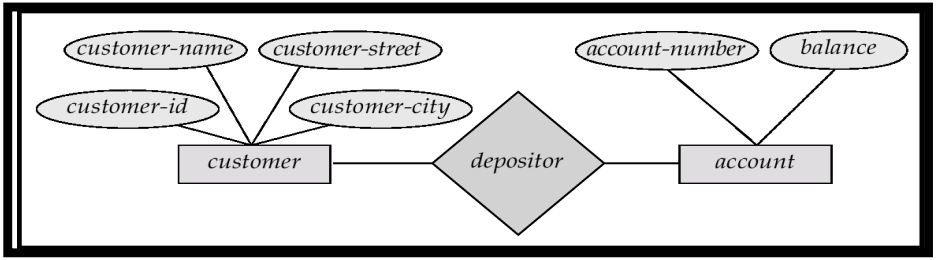
- Constraints are used to define the database structure.
- Constraints typically include elements (and their data types) as well as groups of elements (e.g. entity, record,table), and relationships among such groups.
- Constraints specify some restrictions on valid data; these constraints must be enforced at all times

Data Model Operations

- These operations are used for specifying database retrievals and updates.
- Operations in the data model may include basic model operations and user-defined operations.

Categories of Data Models

- **Conceptual (high-level, semantic) data models**:
  1. Provide concepts that are close to the way many users perceive data.
  2. Also called entity-based or object based data models
  3. Use concepts such as entities, attributes, and relationships.
     - Entities : Represent real world objects or concepts.
     - Attribute : Some property of interest that further describes the entity.
     - Relationship : An association between the entities.



- **Object-Based Data Model** : Extending the E-R model with notions of encapsulation, methods (functions) and object identity.

Categories of Data Models

Physical (low-level, internal) data models:

- Provide concepts that describe details of how data is stored in the computer. (record formats,record orderings and access paths)
- Specified in an ad-hoc manner through DBMS design and administration manuals.
- The concepts provided by physical data models are generally meant for computer specialists, not for end users.

Categories of Data Models

Implementation (representational) data models:

- Provide concepts that fall between the two.
- Models frequently used by traditional DBMSs (Relational data model, Network and hierarchical Models and record-based data models)
- Relational Data model uses a collection of tables to represent both data and the relationships among those data.
- Each table has multiple columns, and each column has unique name

Self-describing data models:

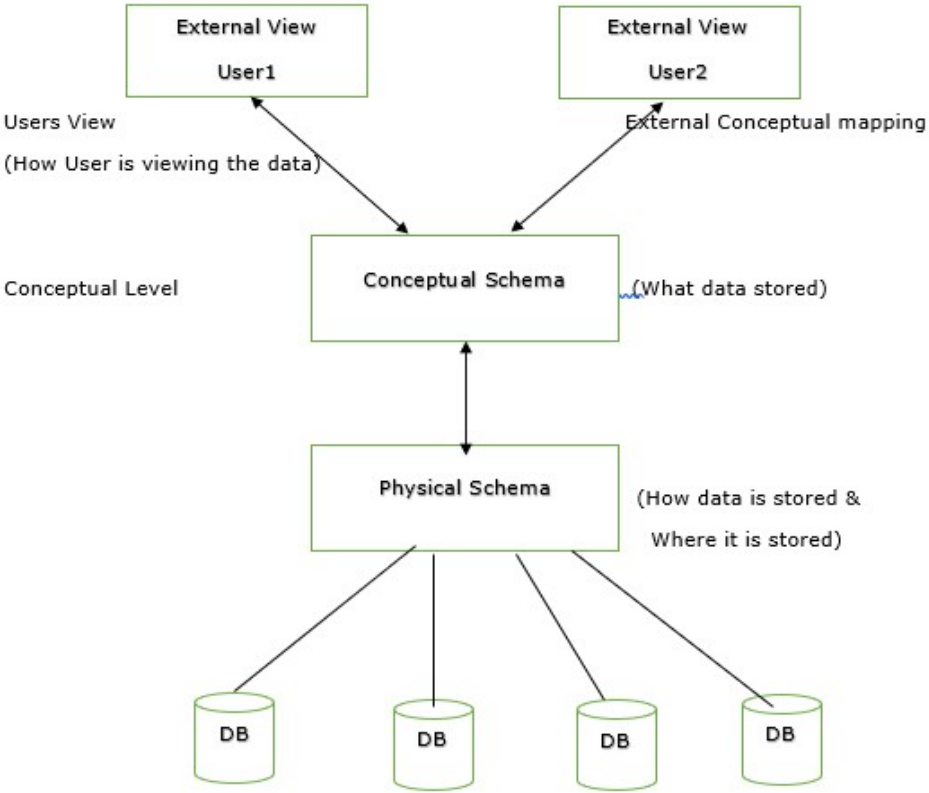
- The data storage in systems based on these models combines the description of the data with the data values themselves.
- These models include XML and NOSQL systems for managing big data

3-schema Architecture

The three schema architecture was proposed to help achieve and visualize the following characteristics of database systems:

- Self-describing nature of data
- Program-data and program-operation independence
- Support of multiple user views.

3-schema Architecture



3-schema Architecture

- **The internal level:**
  - It has an internal schema which describes the physical storage structure of the database.
  - Uses physical data model and describes the complete details of data storage and the access paths for the db
- **The Conceptual level:**
  - The conceptual level has a conceptual schema, which describes the whole database for a community of users.
  - Describes the structure of the db hides details of physical storage structures.
  - Describes entities, data types, relationships, user operations, and constraints
  - A representational data model is used to describe the conceptual schema when database system is implemented.
- **The External level:**
  - Each external schema describes the part of the database that a particular user-group is interested in and hides the rest of the database. Implemented using the representational data model based on the high-level conceptual data model

3-schema Architecture

Three schemas are only descriptions of data. Each user group refers to its own external schema

- The DBMS must transform a request specified on an external schema into a request against the conceptual schema, and then into a request on the internal schema for processing over the stored database.
- If the request is a database retrieval, the data extracted from the stored database must be reformatted to match the user's external view.
- The processes of transforming requests and results between levels are called mappings.

Relational data model

Customer-id	*customer-name*	*customer-street*	*customer-city*	*account-number*
192-83-7465	Johnson	Alma	Palo Alto	A-101
019-28-3746	Smith	North	Rye	A-215
192-83-7465	Johnson	Alma	Palo Alto	A-201
321-12-3123	Jones	Main	Harrison	A-217
019-28-3746	Smith	North	Rye	A-201

Schemas

Schema

- The description of a database is called the database schema
- Specified during database design and is not expected to change frequently.
- A displayed schema is called a **schema diagram**.
- Object in the schema are called as schema construct. Eg: Student,course etc.,
- A schema diagram displays only some aspects of a schema, such as the names of record types and data items, and some types of constraints.

STUDENT

Name	Student_number	Class	Major

COURSE

Course_name	Course_number	Credit_hours	Department

PREREQUISITE

Course_number	Prerequisite_number

Database State or Instance:

- The actual data stored in a database at a particular moment in time - database instance (or occurrence or snapshot).
- The database schema changes very infrequently.
- The database state changes every time the database is updated.

COURSE

Course_name	Course_number	Credit_hours	Department
Intro to Computer Science	CS1310	4	CS
Data Structures	CS3320	4	CS
Discrete Mathematics	MATH2410	3	MATH
Database	CS3380	3	CS

SECTION

Section_identifier	Course_number	Semester	Year	Instructor
85	MATH2410	Fall	04	King
92	CS1310	Fall	04	Anderson
102	CS3320	Spring	05	Knuth
112	MATH2410	Fall	05	Chang
119	CS1310	Fall	05	Anderson
135	CS3380	Fall	05	Stone

GRADE_REPORT

Student_number	Section_identifier	Grade
17	112	B
17	119	C
8	85	A
8	92	A
8	102	B
8	135	A

PREREQUISITE

Course_number	Prerequisite_number
CS3380	CS3320
CS3380	MATH2410
CS3320	CS1310

Relational Model

- The relational Model of Data is based on the concept of a Relation
- A Relation is a mathematical concept based on the ideas of sets
- A relation looks like a table of values.
- A relation typically contains a **set of rows.**
- The data elements in each row represent certain facts, rows are called as **tuples**
- Each column has a column header called as **attribute name**

Relational Model - Example

Relation Name

Attributes

STUDENT

Tuples

Name	Ssn	Home_phone	Address	Office_phone	Age	Gpa
Benjamin Bayer	305-61-2435	373-1616	2918 Bluebonnet Lane	NULL	19	3.21
Chung-cha Kim	381-62-1245	375-4409	125 Kirby Road	NULL	18	2.89
Dick Davidson	422-11-2320	NULL	3452 Elgin Road	749-1253	25	3.53
Rohan Panchal	489-22-1100	376-9821	265 Lark Lane	749-6492	28	3.93
Barbara Benson	533-69-1238	839-8461	7384 Fontana Lane	NULL	19	3.25

Formal Definitions - Relational Schema

The Schema (or description) of a Relation:

- Denoted by $R(A1, A2, ...An)$
- R is the name of the relation
- The attributes of the relation are $A1, A2, ..., An$
- Each attribute has a domain D in the relation schema R.

Formal Definition - Domain

- A domain D is set of atomic values.
- D is called domain of A_i denoted by dom(Ai)
- A domain has a logical definition and also has a data-type or a format defined for it.
- Domain is thus given a name, data type and format
- Degree of a relation is the number attributes n in the relation
- Ex:CUSTOMER (Custid, Custname, Address, Phone) For example, the domain of Custid is 6 digit numbers

Formal Definitions - Tuple

- A tuple is an ordered set of values (enclosed in angled brackets '$< \ldots >'$)
- Each value is derived from an appropriate domain.
- A row in the CUSTOMER relation is a 4 valued tuple and would consist of four values
- ex: $<$ $632895, "JohnSmith", "101MainSt.Atlanta, GA30332", "(404)894-2000" >$

Formal Definitions - Relation State

Given $R(A1, A2, ..., An)$

- The relation r of the relation schema R(A1, A2, ..., An)
- R is the name of the relation
- $A1, A2, \ldots ...An$ are the attributes of the relation
- $r(R)$: a specific state (or "value" or "population") of relation R – this is a set of tuples (rows)
- $r(R) = t1, t2, \ldots, tn$ where each ti is an n-tuple
- $ti =< v1, v2, \ldots, vn >$ where each $v_j element - of dom(Aj)$
- The relation state is a subset of the Cartesian product of the domains of its attributes

$$r(R) \in dom(A1) * dom(A2)....dom(An) \tag{1}$$

Characteristics of Relational

- **Ordering of tuples in a relation:** The tuples are not considered to be ordered, even though they appear to be in the tabular form
- **Ordering of attributes:** We will consider the attributes in R(A1, A2, ..., An) and the values in t = $< v1, v2, ..., vn >$ to be ordered.
- **Values in a tuple:** All values are considered atomic (indivisible). A special null is used to represent unknown or inapplicable values

Relational Model Constraints

There are many restrictions or constraints on the actual values in a database state.

- **Model-based constraints or implicit constraints**: Constraints inherent in the data model Eg: Relation cannot have duplicate tuples (or) no two tuples have the same combination of values
- **Schema-based constraints or explicit constraints:** Constraints directly expressed in the schemas of the data model Specified using DDL statements Eg: Key constraints,Domain constraints etc.,
- **Application-based or semantic constraints or business rules.** Constraints enforced by the application programs Eg: Triggers, views, stored procedures

schema-based constraints

The schema-based constraints include domain constraints, key constraints, constraints on NULLs, entity integrity constraints, and referential integrity constraints.

- **Domain Constraints:** Domain constraints specify that within each tuple, the value of each attribute A must be an atomic value from the domain dom(A).
- Domain constraint violation take place if it does not obey datatype constraints.

Keys

Constraints are conditions that must hold on all valid relation: instances.
Three types of keys:

- Candidate keys
- Primary keys and Alternate keys
- Foreign keys

Candidate Keys

- A relation is defined as a set of tuples, by definition all elements in a set are distinct so all tuples are distinct.
- At any given time, no two tuples in any valid relation instance r(R) will have the same value for the set of attributes.

$$t1[SK] \neq t2[SK] \tag{2}$$

- Any such subsets are called as **superkey**.
- A superkey specifies a uniquness constraint such that no two distinct tuples in any state of r(R) can have same value.
- Every relation should have one default superkey.

Candidate Keys

- A key k of a relation schema R is a superkey of R with an additional property.
- Removing any attribute A from K leaves a set of attributes K+ that is not a superkey of R any more.
- Let K be the set of attributes of R. Then K is the candidate Key or Key for R if it has following properties
  - **Uniqueness:** No value of two distinct tuples of R contains the same value for K
  - **Minimal superkey:** A superkey from which we cannot remove any attributes and still have the uniqueness constraint hold.
  - This minimality property is required for a key but is optional for a superkey.
- A key is a superkey but not vice versa
- A superkey must be key (if it is minimal) or may not be a key

Candidate Keys - Example

- First and Last do not give a unique identifier for each row but in ID uniqueness hold.
- Any set of attributes that includes ID uniquness hold. ID, First, ID, Last and ID, First, Last satisfy uniqueness, but are not minimal.
- Candidate keys are **ID** and **First, Last**

ID	First	Last
S139	John	Smith
S140	Mary	Jones
s141	John	Brown
S142	Jane	Smith

Table 1: Emp_Table

Primary Key

A database has following implicit properties:

- A given relation may have two or more candidate keys.
- Exactly one of those keys be chosen as the primary key, the others are then called alternate keys
- We could choose either ID or First, Last as the Primary Key.
- ID is more convenient as it is a single column and we know it will always be unique (what happens if another John Smith is added?)

Entity Integrity

Entity integrity constraint states that:

- No Primary Key (PK) value can be NULL, in any tuple of r(R)
- This is because primary key values are used to identify the individual tuples in a relation.
- If two or more tuples had NULL for their primary keys we are not able to distinguish them
- Other attributes of R may be constrained to disallow null, even though they are not members of the PK

Foreign Keys

A constraint involving two relations:

- Used to specify a relationship among tuples in two relations: the referencing relation and the referenced relation.
- A foreign key is a set of attributes, say FK, of referencing relation whose values are required to match values of candidate key CK of referenced relation.
- Referential integrity – the database must not contain any unmatched foreign key values
- The value in the foreign key column FK of the referencing relation can be :
  - Same domain value of the referenced relation
  - A value of an existing primary key PK in the referenced relation, or
  - A null

Referential Integrity

DID	DName
13	Marketing
14	Accounts
15	Personnel

Table 2: Dept_Table

DID is a Foreign Key in Employee.
Each Employee's DID value is either NULL, or matches an entry in the Department relation.

ID	Name	DID
15	John Smith	13
16	Mary Jones	14
17	John Brown	13
18	Jane Smith	Null

Table 3: Emp_Table

Referential Integrity

ID	Name	Manager
E1496	John Smith	E1499
E1497	Mary Brown	E1498
E1498	Mark Jones	E1499
E1499	Jane Smith	NULL

Table 4: Emp_Table

- ID is a Candidate Key for Employee, and Manager is a Foreign Key, which refers to the same relation – every tuple's Manager value is either NULL or matches an ID value.
- FOREIGN KEY Manager REFERENCES Employee(ID)

Referential Integrity - violations

- When relations are updated, referential integrity can be violated.
- When a referenced tuple is updated / deleted, an exception will be raised
- To compensate this, the options are:
  - Cascade: Actions cascaded to the matching tuples
  - Restrict: Restricted to the case where there are no matching tuples
  - Nullify: Make values NULL

Referential Actions – An Example

What happens if

- Marketing's DID is changed to 16 in Department?
- The entry for Accounts is deleted from Department?

DID	DName
13	Marketing
14	Accounts
15	Personnel

Table 5: Dept_Table

ID	Name	DID
15	John Smith	13
16	Mary Jones	14
17	John Brown	13
18	Jane Smith	Null

Table 6: Emp_Table

Referential Actions – CASCADE

CASCADE allows the changes made to flow through

- If Marketing's DID is changed to 16 in Department, then the DIDs for John Smith and John Brown also changes
- If Accounts is deleted then so is Mary Jones

DID	DName
16	Marketing
15	Personnel

Table 7: Dept_Table

ID	Name	DID
15	John Smith	16
17	John Brown	16
18	Jane Smith	Null

Table 8: Emp_Table

Referential Actions - Nullify

- NULLIFY sets referenced values to NULL
- If Marketing's DID changes then "John Smith and John Brown " DIDs are set to NULL
- If Accounts is deleted, Mary Brown's DID becomes NULL

DID	DName
13	Marketing
14	Accounts
15	Personnel

Table 9: Dept_Table

ID	Name	DID
15	John Smith	Null
16	Mary Jones	Null
17	John Brown	Null
18	Jane Smith	Null

Table 10: Emp Table

Summary

- Data Abstraction
- Data Models - Types
- Relational Data model
- Keys

Reference

Fundamentals of Database systems 7th Edition by Ramez Elmasri.